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Energy-intense production-inventory planning with participation in sequential energy markets

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HIGHLIGHTS

- Integrated bidding in sequential electricity markets and production scheduling.
- Modeling of information revelation as a multistage stochastic program.
- Comprehensive real-world case study on the cement production process.
- Significant demand-response activation incentives in the German market identified.

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ABSTRACT

To support the uprise of demand response, especially in the context of industrial processes, we propose a new approach to integrally determine the production-inventory plan and the cost-minimizing bids to participate in sequential reserve and energy-only markets. In particular, our approach considers time-coupling constraints which occur in the context of a production-inventory planning problem. We extend this problem with a comprehensive bidding formulation, which allows evaluating revenues and potential cost from the market participation, considering price uncertainties and uncertain activations of committed reserve capacity. This results in a multistage stochastic mixed-integer linear program, which explicitly considers the stage-wise revelation of information in our setup. To illustrate the capabilities of our approach, we apply our model to a real-world case study in which we investigate the participation of a cement plant in the German energy-only and reserve markets. The results of our case study indicate significant revenues for flexible industrial processes when participating in German spot and reserve markets.

1. Introduction

1.1. Motivation

The growth in variable renewable generation in the European electricity sector increases the need for flexibility. Compared to other flexibility options, such as storages, network expansion or flexible power plants, demand response offers a cost-effective way to follow the fluctuating generation. This paper addresses the research challenge to integrate the production-inventory planning with the fully coordinated participation in sequential energy markets. To address market uncertainties associated with prices and activations of reserve, we model the decision process as a multi-stage Stochastic Mixed-Integer Linear Programming (SMILP) which accounts for the stage-wise revelation of information.

1.2. Literature review

The literature on production and inventory planning contains several models considering uncertainties, which are reviewed by Mula et al. [1]. Tang et al. [2] propose a SMILP problem with a focus on energy cost considering non-linear part-load energy consumption of a steel

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Nomenclature

Parameter

Δt	activation time
$\mathbb{E}[\cdot]$	expected value
ϵ	activation duration
γ	reserve price
Ω	sample space
ψ	spot price
ζ	number of activations
a	factor to relate in- and output in a production process
b	factor of energy intensity
c	price
d	demand
g	production
h	weighting factor of energy price
p	power
s	inventory level

Superscripts

α	semi-finished products
β	finished products
Δ	deviation
aux	auxiliary
bo	backorder
curt	curtailment
init	initial
op	operation
pc	product change
p	procured
r, +	positive reserve market
r, −	negative reserve market
re	reserve energy
rt	reserve total price
ru	ramp-up
r	reserve
su	start-up

Variables

C	cost
D	demand placed on backorder or curtailed
G	production in-/output
P	electrical power
R	revenue
S	inventory level
Y	binary variable

Sets

$A(d) \subset T$	subset with hourly time intervals of day d
$d \in D$	set of days
F	sigma algebras
$i \in I$	set of price points for reserve capacity price
$j \in J$	set of scenarios
$k \in K$	set of price points for energy-only bids price
$l \in L$	set of price points for reserve energy price
$m \in M$	set of machines
$n \in N$	set of reserve time slices
$o \in O^{\beta}$	set of finished products
$o' \in O^{\alpha}$	set of semi-finished products
$t \in T$	set of hourly time intervals

production process and uncertain demand. Moon and Park [3] propose a flexible job-shop scheduling problem which considers electricity cost with distributed energy resources and energy storage. The uncertainty of renewable energy in a single-machine scheduling problem is investigated by Liu [4]. Zhang et al. [5] investigate the optimal emission trading strategy within a dynamic multi-technology production-inventory problem. And lately, Golari et al. [6] investigate the production-inventory planning problem of a system which is powered with uncertain on-site and grid renewable energy. These publications address the challenge of scheduling in the context of energy related uncertainties, however uncertainties related with the electricity market participation have not been addressed so far.

The demand response literature investigates how flexible demand can enhance the integration of high shares of renewable energies. On a system perspective, Ambrosius et al. investigate investment incentives for flexible energy consumption offered by the energy-only market [7] and the impact on the system under different market designs [8]. Their findings show that flexible industrial consumption increases welfare considerably. Finn and Fitzpatrick [9] find that industrial demand-side-management can increase the utilization of wind energy in Ireland. From the perspective of independent consumers, Xenos et al. [10] investigate the demand response potential of a process plant. Summerbell et al. [11] study the cost and carbon reduction potentials of a cement plant through load shifting under real-time pricing. Ramin et al. [12] consider a power-intensive production plant and propose two separate Mixed-Integer Linear Programming (MILP) problems to minimize the energy procurement cost in the day-ahead market and maximize the revenues in the reserve market. Wu and Sioshansi [13] propose a two-stage stochastic problem for the scheduling of electric vehicle charging loads under uncertainty and enhance their methodology to include the provision of frequency regulation [14]. Within the demand response literature typically the details associated with the bid formulation to participate in electricity markets are not considered.

The literature on bidding in electricity markets contains one strand which investigates the behavior of independent market participants while the behavior of the remaining bidders is summarized in a probability distribution of the market price, as identified in the review by Li et al. [15]. Early contributions in this context consider bidding strategies in single markets. Wen and David [16] identify optimal bidding strategies for Generation Companies (GenCos) in a day-ahead energy market. Ma et al. [17] extend this model and include risk measures in the objective function. Fleten and Kristoffersen [18] propose a day-ahead bidding model for a Nordic hydro-power producer including both piece-wise linear bids and block bids. From the perspective of an independent large scale consumer, Zare et al. [19] determine the bids to participate in the day-ahead market. A demand-side bidding model that captures uncertain market prices is proposed by Ottesen et al. [20]. Iria et al. [21] identify optimal bids for an aggregator of small prosumers participating in the day-ahead market.

Whenever the participation in subsequent markets is considered when identifying bids in a specific market, the term coordinated bidding is used, see e.g. Boomsma et al. [22] and Ottesen et al. [23]. In energy-only markets, the realistic assessment of the risk of not being dispatched in the first market leads to higher bidding prices in the first market and overall revenue gains, compared to a separate bidding strategy [22]. However, if in two sequential markets balancing power and electrical energy are traded, the two markets can hardly be investigated separately, since the marginal cost of power reservation of a supplier/buyer is directly linked to the producer/consumer surplus in the energy-only market. For detailed information on the economics of balancing power markets the reader if refereed to the work of Müsgens et al. [24]. Typically, coordinated bidding is conducted under

uncertainty and results in a complex decision processes even for independent market participants, this is analyzed in several recent contributions: Kristoffersen [18], Faria and Fleten [25], consider subsequent bidding at the day-ahead and the intra-day market (Elbas Market); Swider [26] and Swider and Weber [27] propose an approach to determine optimal bids for the simultaneous bidding on the day-ahead spot and reserve market; Boomsma et al. [22] propose a model for the coordination of bidding strategies in the day-ahead and the hour-ahead balancing market in the Nordic power markets; Whereas Kumbartzky et al. [28] propose a SMILP to identify the optimal strategy of a combined heat and power plant when submitting energy bids in the German day-ahead market and capacity bids in the reserve market. All these contributions thus provide valuable insights on optimal bids in related markets. To the best of our knowledge, we are the first, however, to incorporate precise modeling of the different aspects of reserve markets, where bidders submit separate capacity- and energy-bids which induce different winning and activation probabilities respectively. This adds additional stages of decisions and information revelation to the problem considered. Regarding the participation of the demand side in multiple markets, Iria et al. extend their work [21] to investigate aggregated prosumers participating in the reserve and day-ahead market [29]. Ottesen et al. [23] propose bidding models for demand-side flexibility aggregators in the balancing and day-ahead market. Due to the complexities arising in the context of load aggregation, these works model the market participation and the electricity consuming process in a simplified manner. In contrast to these works, we incorporate the daily update of information considering the market prices and activation of reserve. In combination with a detailed production-inventory model, this allows us to capture inter-temporal dependencies throughout the entire optimization horizon and, thus, identify feasible and cost optimal bids to participate in the respective markets.

1.3. Contribution of this work

We propose a SMILP problem to integrally plan production-inventory and market participation in sequential energy markets. This contributes to two strands of literature. First, our model extends the existing literature on stochastic production-inventory models by the integration of a detailed market formulation that enables the planner to determine not only the production schedule, but also the bids to participate in sequential energy markets under price uncertainty and uncertain activations of committed reserve capacity. Second, and most importantly, we contribute to the recently developing literature which investigates bidding strategies in multiple electricity markets from a consumers perspective. In contrast to the existing literature, we investigate the participation in day-ahead reserve and energy markets in combination with a underlying production process. The identification of cost optimal bids to participate in the respective markets while identifying feasible production schedules facing uncertain activations of reserve requires (i) the explicit consideration of the relevant details regarding reserve markets (capacity- and energy-bids, as well as resulting activation probabilities), (ii) a market formulation that captures all stages of information revelation to cover time interdependences throughout several days and (iii) a detailed production-inventory model.

1.4. Paper organization

The remainder of this paper is organized as follows. First, we explain the markets and the sequence of decisions market participants face. This is followed by the presentation of our model. We conduct a real-world case study in which we apply the framework to a cement plant in the German market environment. We conclude with a summary of our findings and identify future research questions.

2. Energy-only and reserve markets

To ensure the system balance at any point in time, a sequence of energy-only and reserve markets is established in liberalized electricity markets.

Day-ahead energy-only markets are typically characterized by double-sided auctions in which market participants trade electrical energy. One day prior to production, producers, as well as consumers, submit hourly bids which consist of the volume they want to sell/buy and a price at which they are willing to sell/buy. Evaluation of the aggregated bids of supply and demand result in the market clearing price which is determined by the highest accepted supply bid (uniform pricing).

Reserve markets are typically characterized by single buyer auctions, where the Transmission System Operator (TSO) procures reserves. Producers and consumers are remunerated for the reservation of upward or downward regulation capacity and, in case of activation, also for the actual delivery of energy. If at some point in time electricity generation (consumption) exceeds consumption (generation) the frequency rises (drops) and beforehand procured negative (positive) balancing power is activated by the TSO. Typically, the TSOs procure different types of control reserve, which differ according to their speed of activation. Due to the technical requirements of the different types of balancing power, industrial companies are usually able to provide Frequency Restoration Reserve (FRR) or Replacement Reserve (RR).

The Market Sequence determines the decision and information structure market participants face. Typically, reserve capacity and energy-only auctions are conducted on the day ahead of delivery,[1] where reserve and energy-only markets are conducted sequentially in a market specific sequence.[2] Our work is applicable in sequential markets in which the reserve capacity auction is followed by the energy-only auctions as e.g. in Germany [31], Austria [32], Denmark [33] or in which the energy-only auction is followed by the reserve auction as e.g. in Spain [34] or Switzerland [35]. For didactic reasons, our illustration in Fig. 1 and our case study in Section 4 focus on a market sequence in which the energy-only auction is conducted subsequently to the reserve auction. Throughout the article, we outline the adaptations necessary to capture the reverse sequence of energy and reserve markets (see Section 3.6).

After the reserve and energy-only auctions are conducted the final production and consumption timetables are determined. Within real time the TSO monitors the grid frequency and activates beforehand procured reserve capacities to balance any deviation. Reserves are activated according to a merit order, and market participants are granted a specific activation time to ramp-up or ramp-down their reserved capacities.

3. Methodology

3.1. Production and bidding planning of an independent industrial company in sequential energy markets

We consider an independent industrial company actively participating in sequential reserve and energy-only markets. Bids to participate in the different markets as well as production-inventory plans are determined under uncertainty, the respective stochastic processes are described in Section 3.2. The company identifies reserve bids and energy-only bids which are subject to different market rules, these are

[1] Ocker et al. [30] investigate the European balancing power markets in 24 countries and identify 25 markets with daily auctions and 16 markets with weekly auctions.

[2] The basic idea of our work is also applicable in markets with integrated, "co-optimized" procurement of reserve, as e.g. in the USA. However, the detailed mathematical formulation of these market rules is beyond the scope of this work.

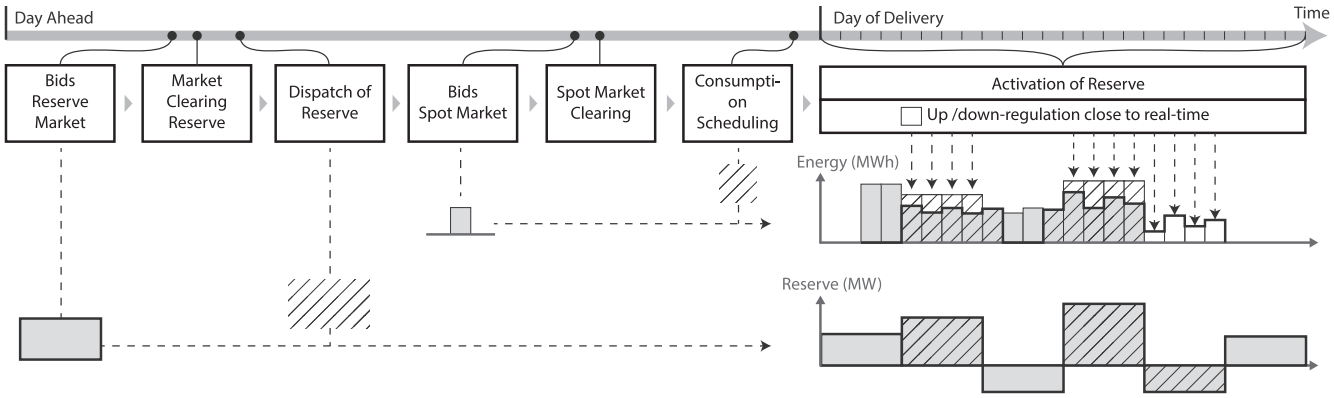


Fig. 1. Sequence of short-term electricity markets in case spot market trading takes place subsequently to reserve market trading, and corresponding dispatch of capacity and production scheduling of an exemplary industrial market participant.

modeled in Sections 3.5 and 3.4, respectively. The market participation in sequential reserve and energy-only markets requires a coordinated bidding strategy anticipating outcomes on future markets, this leads to a complex decision sequence even for independent market participants (compare Section 1.2). We consider a dynamic production-inventory system, which is subject to time-coupling constraints. In this context it is of vital importance to capture dynamic effects throughout subsequent days under different realizations of uncertain parameters to ensure reliability of costumer supply. We ensure this with a detailed modeling of the decision sequence which is presented in Section 3.6 in combination with a detailed modeling of the underlying technical process which is presented in Section 3.3. We moreover explicitly take into account stochastic parameters describing the activation of reserve (compare Section 3.2), which further increase the complexity of the sequence of the multistage decision problem of the considered firm (compare Section 3.6).

3.2. The stochastic process

Our model considers an optimization horizon of $d \in D := \{1, ..., |D|\}$ days with hourly time intervals $t \in T_1 := \{V + 1, ..., V + 24|D|\}$. We further consider a time horizon of V hours with hourly time intervals $t \in T_2 := \{1, ..., V\}$ of uncertain realizations of reserve activation. Together this builds the entire considered horizon $T = T_2 \cup T_1$. We further introduce the subsets $A(d) \subset T_1$ with $t \in A(d) := \{24(d-1) + 1 + V, ..., 24d + V\}$ which include hourly time intervals of each day d. Adopting the widely used perspective of price-taking behavior of bidders, we analyze the behavior of one single bidder explicitly, whereas the behavior of the remaining bidders is summarized by a stochastic process regarding market prices.[3] Since the spot and reserve market prices for the time horizon $t \in T_2$ are already revealed, we introduce the stochastic processes of uncertain spot prices $\{\psi_t\}_{t\in T_1}$, positive power reserve prices $\{\gamma_t^{r,+}\}_{t\in T_1}$ as well as negative power reserve prices $\{\gamma_t^{r,-}\}_{t\in T_1}$. The duration of activations are unknown for the entire time horizon $T = T_2 \cup T_1$. To describe the procedure of activations we introduce the stochastic processes of activation duration $\{\epsilon_t^{\text{called}\pm}\}_{t\in T}$ and number of activations $\{\zeta_t^{\text{called}\pm}\}_{t\in T}$ within each time interval t. Since a bid is activated according to its position in the merit order, the stochastic variables are dependent on the energy bid price c^{re}. We use discrete energy bid prices c_l^{re} with $l \in L := \{1, ..., |L|\}$ price points and the resulting stochastic processes are $\{\epsilon_{t,l}^{\text{called}\pm}\}_{t\in T}$ and $\{\zeta_{t,l}^{\text{called}\pm}\}_{t\in T}$. The stochastic processes are defined on some filtered probability space (Ω, F, P) where Ω is the sample space, F are the sigma-algebras and P is the probability measure. The filtration describes how information is revealed over the considered time horizon and is dependent on the sequence of markets. For a daily sequence of reserve and energy-only markets the sigma algebras are $F_1 \subseteq F_2 \subseteq F_3 \subseteq ... \subseteq F_{3d-2} \subseteq F_{3d-1} \subseteq F_{3d} \subseteq F_{3d+1}$. The information of each day $d \in 1, ..., D$ before and after market clearing of the reserve market is given by F_{3d-2} and F_{3d-1}, the information before and after energy-only market clearing is given by F_{3d-1} and F_{3d} and the information before and after activation of reserve is given by F_{3d} and F_{3d+1}. For computational reasons, we represent the stochastic process by a finite number of scenarios j for discrete price levels l. This yields the stochastic process of market prices $\left\{\psi_{t,j}, \gamma_{t,j}^{r,\pm}\right\}_{t\in T_1}, j \in J := \left\{1, ..., \left|J\right|\right\}$ and activations $\left\{\epsilon_{t,j,l}^{\text{called}\pm}, \zeta_{t,j,l}^{\text{called}\pm}\right\}_{t\in T}, j \in J := \{1, ..., |J|\}, l \in L := \{1, ..., |L|\}$. The probability of each scenario is given by w_j. The scenario sampling as well as the generation and reduction are detailed in Sections 4.2 and 4.3, respectively.

3.3. Modeling of the production process

We consider the production of finished products $o \in O^{\beta}$. Products are produced on multiple machines $m \in M$. The output of each machine is given by $G_{t,m,j,o}^{\beta}$. Each finished product consists of one or multiple semi-finished products $o' \in O^{\alpha}$ which enter the machine as input $G_{t,m,j,o'}^{\alpha}$. The relation of input and output is then given as:

$$G_{t,m,j,o}^{\beta} = a_{o,o'} G_{t,m,j,o'}^{\alpha} \tag{1}$$

The energy intensity of product o is given by the parameter b_o. The power consumption $P_{t,m,j,o}$ of machine m is therefore modeled as a linear function of the product output:

$$P_{t,m,j,o} = b_o G_{t,m,j,o}^{\beta} \tag{2}$$

Non-linear correlations of in- and output could be modeled by piecewise linear approximations, as e.g. in [36]. The binary decision variable $Y_{t,m,j,o}$ indicates whether product o is produced in time interval t on machine m in scenario j or not. The energy consumption is limited by the upper bound $p_m^{\max}$.

$$P_{t,m,j,o} \leqslant Y_{t,m,j,o} p_m^{\max} \tag{3}$$

During each time interval one product is assigned to each machine, regardless of the actual state of operation.

$$\sum_{o\in O^{\beta}} Y_{t,m,j,o} = 1 \tag{4}$$

The energy consumption to produce products o on machine m in time interval t equals the energy procured on the markets $P_{t,m,j}^{p}$ for

[3] This assumption is widely used in the literature (e.g. [22–34]) and justified by the fact that the bidding capacities are relatively small compared to the overall market volume.

machine m.

$$\sum_{o\in O^{\beta}} P_{t,m,j,o} = P^{\mathrm{p}}_{t,m,j} \tag{5}$$

The binary decision variable $Y^{\mathrm{op}}_{t,m,j}$ indicates whether machine m is in operation during time interval t and in scenario j or not. The energy consumption of each machine is bounded by its capacity limits.

$$Y^{\mathrm{op}}_{t,m,j} p^{\min}_{m} \leqslant P^{\mathrm{p}}_{t,m,j} \leqslant Y^{\mathrm{op}}_{t,m,j} p^{\max}_{m} \tag{6}$$

The binary decision variable $Y^{\mathrm{pc}}_{t,m,j,o}$ indicates whether the product on machine m has been changed during time period t or not.

$$Y^{\mathrm{pc}}_{t,m,j,o} \geqslant Y_{t,m,j,o} - Y_{t-1,m,j,o} \quad \forall\, t \in T\backslash\{1\} \tag{7}$$

The binary decision variable $Y^{\mathrm{su}}_{t,m,j}$ indicates whether machine m is starting up in time period t and scenario j or not.

$$Y^{\mathrm{su}}_{t,m,j} \geqslant Y^{\mathrm{op}}_{t,m,j} - Y^{\mathrm{op}}_{t-1,m,j} \quad \forall\, t \in T\backslash\{1\} \tag{8}$$

The primary task of the production process is to meet customers demand $d^{\beta}_{t,o}$ for product o. We consider inventory capacities for each finished product $o \in O^{\beta}$ with inventory level $S^{\beta}_{t,j,o}$.

$$d^{\beta}_{t,o} + D^{\beta,\mathrm{bo}}_{t-1,j,o} = S^{\beta}_{t-1,j,o} - S^{\beta}_{t,j,o} + \sum_{m\in M} G^{\beta}_{t,m,j,o} + D^{\beta,\mathrm{curt}}_{t,j,o} + D^{\beta,\mathrm{bo}}_{t,j,o} \quad \forall\, t \in T\backslash\{1\} \tag{9}$$

$$d^{\beta}_{t,o} \geqslant s^{\mathrm{init},\beta}_{o} - S^{\beta}_{t,j,o} + \sum_{m\in M} G^{\beta}_{t,m,j,o} + D^{\beta,\mathrm{curt}}_{t,j,o} + D^{\beta,\mathrm{bo}}_{t,j,o} \quad t \in \{1\} \tag{10}$$

If inventory capacities and production are not sufficient to meet customers demand, demand can be delayed to the next period $D^{\beta,\mathrm{bo}}_{t,j,o}$ or curtailed by a specific amount $D^{\beta,\mathrm{curt}}_{t,j,o}$. Initial inventory conditions $s^{\mathrm{init},\beta}_{o}$ define the inventory conditions at the start of the considered horizon.

The parameter $s^{\mathrm{end},\beta}_{o}$ defines the aspired inventory level in the last time interval, which offers flexibility for future production optimization. The variables $\Delta S^{\beta+}_{j,o}$ and $\Delta S^{\beta-}_{j,o}$ denote positive and negative deviations of the actual from the aspired inventory level:

$$S^{\beta}_{|T|,j,o} = s^{\mathrm{end},\beta}_{o} + \Delta S^{\beta-}_{j,o} - \Delta S^{\beta+}_{j,o} \tag{11}$$

The inventory capacity is bounded by upper and lower bounds $s^{\mathrm{cap,min},\beta}_{o}$ and $s^{\mathrm{cap,max},\beta}_{o}$, respectively:

$$s^{\min,\beta}_{o} \leqslant S^{\beta}_{t,j,o} \leqslant s^{\max,\beta}_{o} \tag{12}$$

The inventory of semi-finished products is modeled analogously with the difference that the storage input $g^{\alpha}_{t,o}$ is given exogenously and the discharge output is determined endogenously. Moreover, demand of semi-finished products can neither be curtailed nor shifted to later periods.

$$\begin{aligned}
&\sum_{m\in M} G^{\alpha}_{t,m,j,o} = S^{\alpha}_{t-1,j,o} - S^{\alpha}_{t,j,o} + g^{\alpha}_{t,o} \quad \forall\, t \in T\backslash\{1\}\\
&\sum_{m\in M} G^{\alpha}_{t,m,j,o} = s^{\mathrm{init},\alpha}_{o} - S^{\alpha}_{t,j,o} + g^{\alpha}_{t,o} \quad t \in \{1\}\\
&S^{\alpha}_{|T|,j,o} = s^{\mathrm{init},\alpha}_{o} + \Delta S^{\alpha-}_{j,o} - \Delta S^{\alpha+}_{j,o}\\
&s^{\min,\alpha}_{o} \leqslant S^{\alpha}_{t,j,o} \leqslant s^{\max,\alpha}_{o}
\end{aligned} \tag{13}$$

3.4. Modeling the participation in the day-ahead market

To model the participation in the day-ahead auction, we adapt a formulation that has been widely used (e.g. [18,25,22,28]). We introduce discrete bidding prices c^{s}_{k} with $k \in K := \{1, ..., |K|\}$ being the index of the discrete price points. We denote the bidding capacity by $\widetilde{P}_{t,j,k}$ and the accepted capacity by $\widehat{P}_{t,j}$. The accepted capacity in scenario j is then defined by:

$$\widehat{P}_{t,j} = \widetilde{P}_{t,j,k} \quad \text{if} \quad c^{s}_{k-1} \leqslant \psi_{t,j} \leqslant c^{s}_{k} \quad \forall\, k \in K\backslash\{1\} \tag{14}$$

We further restrict the bidding curve to be decreasing with increasing bidding price:

$$\widetilde{P}_{t,j,k+1} \leqslant \widetilde{P}_{t,j,k} \quad \forall\, k \in K\backslash\{|K|\} \tag{15}$$

Given the accepted capacity, the energy demand of all machines m must be met:

$$\sum_{m\in M} P^{\mathrm{p}}_{t,m,j} = \widehat{P}_{t,j} \tag{16}$$

Finally, the production costs of each time interval t and each scenario j are calculated as the sum of start-up cost, energy procurement cost, inventory deviation cost as well as demand curtailment and back order cost. All cost components are evaluated using cost factors c, which are set as model parameters by the planner. Unit holding cost could be integrated by penalizing the inventory levels $S_{t,j,o}$ in the objective function.

$$\begin{aligned}
C_{t,j} = &\sum_{m\in M} \left(Y^{\mathrm{su}}_{t,m,j} c^{\mathrm{su}} \right) + \widehat{P}_{t,j}\psi_{t,j} + \sum_{o\in O^{\alpha}} \Delta S^{\alpha+}_{j,o} c^{\Delta\alpha+}_{o} + \sum_{o\in O^{\alpha}} \Delta S^{\alpha+}_{j,o} c^{\Delta\alpha+}_{o} + \sum_{o\in O^{\beta}} \Delta S^{\beta+}_{j,o} c^{\Delta\beta+}_{o} +\\
&\sum_{o\in O^{\beta}} \Delta S^{\beta-}_{j,o} c^{\Delta\beta-}_{o} + \sum_{o\in O} D^{\beta,\mathrm{curt}}_{t,j,o} c^{\beta,\mathrm{curt}}_{o} + \sum_{o\in O} D^{\beta,\mathrm{bo}}_{t,j,o} c^{\beta,\mathrm{bo}}_{o}
\end{aligned} \tag{17}$$

3.5. Modeling of participation in the reserve market

To model the participation in the reserve market, we choose a flexible model formulation to cover different market characteristics, such as permitted activation times, minimum bid sizes, duration of time slices, auction frequencies, number of delivery time slices and their duration as well as different pricing and scoring rules.

First, we introduce discrete power bidding prices $c^{r,-}_{t,i}$ and $c^{r,+}_{t,i}$ with $i \in I$ being the index of a possible bid price. Together with the discrete energy bidding prices $c^{re,+/-}_{l}$, this allows us to keep the formulation linear. The quantity of negative reserve power offered at each price point is then denoted by $P^{\mathrm{r},-}_{t,m,j,i,l}$.

The bid quantity of negative reserve is limited by the planned operation point, resulting from the participation in the spot market:

$$\begin{aligned}
P^{\mathrm{r},-}_{t,m,j,i,l} \leqslant\ & p^{\max}_{m} - \sum_{o\in O^{\beta}} P_{t,m,j,o}\\
P^{\mathrm{r},-}_{t,m,j,i,l} \geqslant\ & p^{\min}_{m} - \sum_{o\in O^{\beta}} P_{t,m,j,o} - \left(1 - Y^{\mathrm{r},-}_{t,m,j,i,l}\right) p^{\max}_{m}
\end{aligned} \tag{18}$$

Furthermore, the quantity of the bid is limited by the ramp-up rate of the machine m, and the activation time Δt^{ru} of the considered reserve market:

$$P^{\mathrm{r},-}_{t,m,j,i,l} \leqslant \Delta p^{\mathrm{ru}}_{m} \Delta t^{\mathrm{ru}} \tag{19}$$

Additional bounds of the bid quantity are given by the minimum bid size $p^{\mathrm{r,min}}$ and the maximum bid size $p^{\mathrm{r,max}}$ of the considered reserve market:

$$Y^{\mathrm{r},-}_{t,m,j,i,l} p^{\mathrm{r,min}} \leqslant P^{\mathrm{r},-}_{t,m,j,i,l} \leqslant Y^{\mathrm{r},-}_{t,m,j,i,l} p^{\mathrm{r,max}} \tag{20}$$

The binary decision variable $Y^{\mathrm{r},-}_{t,m,j,i,l}$ indicates whether a bid is submitted at time t for machine m, in scenario j at prices $c^{r,-}_{t,i}$ and $c^{\mathrm{re},-}_{l}$. In some cases, the duration of a reserve time slice consists of multiple day-ahead time slices. We introduce time slices $n \in N := \{1, ..., |N|\}$, with $|N|$ being the number of reserve time slices that are traded within the time horizon T_1. The subsets $A(n) \subseteq T_1$ denote hourly time intervals $t \in A(n)$ that belong to one reserve time slice n. This allows us to model reserve time windows which consist of successive day-ahead time slices (e.g. 4 h time slices) as well as reserve time windows which consist of disjunctive day-ahead time slices (e.g. peak-/off-peak-periods over multiple days).

$$P^{\mathrm{r},-}_{t_1,m,j,i,l} = P^{\mathrm{r},-}_{t_2,m,j,i,l} \quad \forall\, t_1, t_2 \in A(n) \tag{21}$$

We assume that for each machine m only one bid is submitted within each scenario j:

$$\sum_{l\in L}\sum_{i\in I} Y^{\mathrm{r},-}_{t,m,j,i,l} \leqslant 1 \tag{22}$$

The constraints on the bids in the positive reserve market are modeled similarly. First, the quantity of positive reserve offered by a machine m is denoted by $P^{\mathrm{r},+}_{t,m,j,i,l}$ and limited by the current state of operation, the ramp-down rate of the machine as well as by the minimum and maximum bid size:

$$\begin{aligned} P^{\mathrm{r},+}_{t,m,j,i,l} &\leqslant \sum_{o\in O^{\beta}} P_{t,m,j,o} - p^{\min}_{m} Y^{\mathrm{aux}}_{t,m,j,i,l} \\ P^{\mathrm{r},+}_{t,m,j,i,l} &\geqslant \sum_{o\in O^{\beta}} P_{t,m,j,o} - p^{\max}_{m} Y^{\mathrm{aux}}_{t,m,j,i,l} \end{aligned} \tag{23}$$

$$\begin{aligned} &P^{\mathrm{r},+}_{t,m,j,i,l} \leqslant \Delta p^{\mathrm{rd}}_{m} \Delta t^{\mathrm{ru}} \\ &Y^{\mathrm{r},+}_{t,m,j,i,l} p^{\mathrm{r,min}} \leqslant P^{\mathrm{r},+}_{t,m,j,i,l} \leqslant Y^{\mathrm{r},+}_{t,m,j,i,l} p^{\mathrm{r,max}} \end{aligned} \tag{24}$$

We introduce the binary variable $Y^{\mathrm{aux}}_{t,m,j,i,l}$ to ensure that the machine is either ramped down within its capacity limits or it is entirely shut down. The binary decision variable $Y^{\mathrm{r},+}_{t,m,j,i,l}$ indicates whether reserve is offered at price i in scenario j or not. Furthermore, the offered quantity is constant over the duration of one reserve time slice and only one bid per machine is submitted within each scenario.

$$\begin{aligned} &P^{\mathrm{r},+}_{t_1,m,j,i,l} = P^{\mathrm{r},+}_{t_2,m,j,i,l} \quad \forall\, t_1, t_2 \in A(n) \\ &\sum_{l\in L}\sum_{i\in I} Y^{\mathrm{r},+}_{t,m,j,i,l} \leqslant 1 \end{aligned} \tag{25}$$

The most prominent scoring rules in reserve markets are the evaluation of the bids based on the capacity price only and the evaluation of the bids as a w eighted sum of capacity price and energy price[4]. In the latter case, the weighting factor or the energy bid price h is typically set to the average call probability of accepted capacity bids in a preceding trading horizon (e.g. the preceding year). We assume that the bids are fully accepted or rejected. We denote the accepted capacity by $\widehat{P}^{\mathrm{r},-}_{t,m,j,i,l}$ and $\widehat{P}^{\mathrm{r},+}_{t,m,j,i,l}$, respectively. The offered capacity at price pair i, l is accepted if the total scoring price $c^{\mathrm{rt},\pm}_{t,i,l} = c^{\mathrm{r},\pm}_{t,i} + hc^{\mathrm{re},\pm}_{t,l}$ is lower or equal to the marginal market scoring price of negative/positive reserve $\gamma^{\mathrm{r},-}_{t,j}/\gamma^{\mathrm{r},+}_{t,j}$ in scenario j:

$$\widehat{P}^{\mathrm{r},-}_{t,m,j,i,l} = \begin{cases} P^{\mathrm{r},-}_{t,m,j,i,l} & \text{if } c^{\mathrm{r},-}_{t,i} + hc^{\mathrm{re},-}_{t,l} \leqslant \gamma^{\mathrm{r},-}_{t,j} \\ 0 & \text{if } c^{\mathrm{r},-}_{t,i} + hc^{re,-}_{t,l} > \gamma^{\mathrm{r},-}_{t,j} \end{cases}$$

$$\widehat{P}^{\mathrm{r},+}_{t,m,j,i,l} = \begin{cases} P^{\mathrm{r},+}_{t,m,j,i,l} & \text{if } c^{\mathrm{r},+}_{t,i} + hc^{\mathrm{re},+}_{t,l} \leqslant \gamma^{\mathrm{r},+}_{t,j} \\ 0 & \text{if} c^{\mathrm{r},+}_{t,i} + hc^{\mathrm{re},+}_{t,l} > \gamma^{\mathrm{r},+}_{t,j} \end{cases} \tag{26}$$

As mentioned earlier, the activation of reserve energy is executed in real-time.

Consequently, it is not possible to model one specific energy price for one time interval nor one specific electrical power that is called. We therefore model the expected activation duration of a specific bid $\epsilon^{\mathrm{called}+/-}_{t,j,l}$, for a detailed description see Section 4.2. Considering the accepted power capacity $\widehat{P}^{\mathrm{r},+}_{t,m,j,i,l}$ the expected positive energy provided can be calculated as:

$$\widehat{P}^{\mathrm{re},+}_{t,m,j} = \sum_{l\in L, i\in I} \epsilon^{\mathrm{called}+}_{t,j,l}\, \widehat{P}^{\mathrm{r},+}_{t,m,j,i,l} \tag{27}$$

The expected negative reserve energy provided is calculated analogously:

$$\widehat{P}^{\mathrm{re},-}_{t,m,j} = \sum_{l\in L, i\in I} \epsilon^{\mathrm{called}-}_{t,j,l}\, \widehat{P}^{\mathrm{r},-}_{t,m,j,i,l} \tag{28}$$

[4] According to Ocker et al. [30] reserve bids evaluation based on the capacity price is currently implemented e.g. in Austria, Czech Republic and Netherlands/ evaluation based on weighted sum is implemented e.g. in Iceland, Lithuania and Romania.

The adapted energy consumption of the machines is defined as:

$$\sum_{o\in O^{\beta}} P_{t,m,j,o} = \widehat{P}_{t,m,j} + \widehat{P}^{\mathrm{re},-}_{t,m,j} - \widehat{P}^{\mathrm{re},+}_{t,m,j} \tag{29}$$

The revenues from the participation in the reserve markets, in case of a pay-as-bid auction are given by

$$\begin{aligned} R_{t,j} &= \sum_{m\in M}\sum_{i\in I}\sum_{l\in L} (\mathrm{N}/24\mathrm{D}(\widehat{P}^{\mathrm{r},+}_{t,m,j,i,l} c^{\mathrm{r},+}_{t,i} + \widehat{P}^{\mathrm{r},-}_{t,m,j,i,l} c^{\mathrm{r},-}_{t,i}) + \epsilon^{\mathrm{called}-}_{t,j,l}\, c^{\mathrm{re},-}_{l} \widehat{P}^{\mathrm{r},-}_{t,m,j,i,l} \\ &\quad + \epsilon^{\mathrm{called}+}_{t,j,l}\, c^{\mathrm{re},+}_{l} \widehat{P}^{\mathrm{r},+}_{t,m,j,i,l} - \left(Y^{\mathrm{r},-}_{t,m,j,i,l} \zeta^{\mathrm{called}-}_{t,j,l} + Y^{\mathrm{r},+}_{t,m,j,i,l} \zeta^{\mathrm{called}+}_{t,j,l} \right) c_{\mathrm{su}}). \end{aligned} \tag{30}$$

The first term of Eq. (30) refers to the revenues from the sole reservation of reserve, the second term reflects the revenues from the activations of reserve and the last term includes additional cost that occur with every activation. If reserved capacity is remunerated according to uniform pricing the bidding prices $c^{r,\pm}_{t,i}$ need to be replaced by realized market prices. If activated reserve is remunerated according to uniform pricing the bidding price $c^{re,+}_{l}$ needs to replaced by average market price during activation.

3.6. The stochastic programming problem

In the following, the multistage SMILP problem resulting from the stage-wise revelation of information is presented. The multi-stage decision structure is further depicted in Fig. 1. The first stage of each day consists of reserve market bidding with the objective to maximize the expected revenue of bidding in the reserve market resulting from actual reserve market dispatch and bidding in the spot market

$$\max\{\mathbb{E}[R_d(P^{\mathrm{r},+}_t, P^{\mathrm{r},-}_t)|F_{3d-2}]:(19-22,\, 24,\, 25)\},\ t\in A(d). \tag{31}$$

In response to the price realization in the reserve market, in the second stage of each day the bids to participate in the spot market are defined to minimize the expected energy procurement and production cost

$$\begin{aligned} &R_d\left(P^{\mathrm{r},+}_t, P^{\mathrm{r},-}_t\right) \\ &= \max\left\{ \sum_t \mathrm{N}/24\mathrm{D}(\widehat{P}^{\mathrm{r},+}_t c^{\mathrm{r},+}_t + \widehat{P}^{\mathrm{r},-}_t c^{\mathrm{r},-}_t) - \mathbb{E}\left[C^{\mathrm{spot}}\left(\widetilde{P}_t, \widehat{P}^{\mathrm{r},+}_t, \widehat{P}^{\mathrm{r},-}_t\right) \middle| F_{3d-1} \right] : \left(15,\, 18,\, 23,\, 26\right) \right\},\ t\in A(d). \end{aligned} \tag{32}$$

After price realization in the spot market, in the third stage of each day the production plan is determined minimizing the expected production cost anticipating expected activations of reserve.

$$\begin{aligned} &C^{\mathrm{spot}}_d\left(\widetilde{P}_t, \widehat{P}^{\mathrm{r},+}_t, \widehat{P}^{\mathrm{r},-}_t\right) \\ &= \min\left\{ \sum_t \widehat{P}_t \psi_t + \mathbb{E}\left[C^{\mathrm{prod}}\left(\widehat{P}_t, P^{\mathrm{p}}_{t,m}, Y_{t,m,o}\right) \middle| F_{3d} \right] : \left(1-13,\, 16\right) \right\},\ t\in A(d). \end{aligned} \tag{33}$$

After the revelation of activations, the final state of operation is determined and the production cost is minimized considering expected revenues and cost of market participation in the subsequent day.

$$C_d^{\text{prod}}\left(\widehat{P}_t,\, P_{t,m}^{\text{p}},\, Y_{t,m,o}\right)$$
$$= \min\left\{\sum_t \left(Y_{t,m}^{\text{su}} c^{\text{su}} + \sum_{o\in O} D_{t,o}^{\beta,\text{curt}} c_o^{\beta,\text{curt}} + \sum_{o\in O} D_{t,o}^{\beta,\text{bo}} c_o^{\beta,\text{bo}} + \epsilon_t^{\text{called}-} c^{\text{re},-} \widehat{P}_{t,m}^{\text{r},-}\right.\right.$$
$$\left.\left. + \epsilon_t^{\text{called}+} c^{\text{re},+} \widehat{P}_{t,m}^{\text{r},+} - \left(Y_{t,m,\gamma_t}^{\text{r},-,\text{called}-} + Y_{t,m,\gamma_t}^{\text{r},+,\text{called}+}\right) c_{\text{su}}\right) - \mathbb{E}\left[R_{d+1}\left(P_{t_2}^{\text{r},+},\, P_{t_2}^{\text{r},-}\right)\middle| F_{3d+1}\right]\right.$$
$$\left.:(27-29)\right\}, \quad t \in A(d),\, t_2 \in A\left(d+1\right) \tag{34}$$

For the last day $d = |D|$ the term of expected revenues in Eq. (34) is replaced by the storage deviation cost

$$C^{\text{stodev}} = \sum_{o\in O^\alpha} \Delta S_{j,o}^{\alpha+} c_o^{\Delta\alpha+} + \sum_{o\in O^\alpha} \Delta S_{j,o}^{\alpha+} c_o^{\Delta\alpha+} + \sum_{o\in O^\beta} \Delta S_{j,o}^{\beta+} c_o^{\Delta\beta+} + \sum_{o\in O^\beta} \Delta S_{j,o}^{\beta-} c_o^{\Delta\beta-}. \tag{35}$$

Since we consider a discrete distribution of the uncertain parameters, the problem can be equivalently reformulated as the so-called deterministic equivalent, which is presented in Eq. (37), see e.g. [37]. In the case of a linear objective function and linear constraints, the problem can be solved to global optimality and the minimal cost in each stage can be determined.

$$\min \sum_{j\in J} w_j \sum_{t\in T_1} \left(C_{t,j} - R_{t,j}\right) \tag{36a}$$

$$\text{s. t. } (1) - (27) \tag{36b}$$

$$Y_{t,m,j,o},\, Y_{t,m,j}^{\text{op}},\, Y_{t,m,j,o}^{\text{pc}},\, Y_{t,m,j}^{\text{su}},\, Y_{t,m,j,i,l}^{\text{r},+},\, Y_{t,m,j,i,l}^{\text{r},-},\, Y_{t,m,j,o}^{\text{aux}} \in \{0,\,1\} \tag{36c}$$

$$\widehat{P}_{t,m,j,i,l}^{\text{re},-},\, \widehat{P}_{t,m,j,i,l}^{\text{re},+},\, \widehat{P}_{t,m,j,i,l}^{\text{r},-},\, P_{t,m,j,i,l}^{\text{r},-},\, \widehat{P}_{t,m,j,i,l}^{\text{r},+},\, P_{t,m,j,i,l}^{\text{r},+},\, \widehat{P}_{t,m,j,o},\, \widetilde{P}_{t,m,j,o},\, P_{t,m,j,o},\, P_{t,m,j}^{\text{p}}$$
$$D_{t,j,o}^{\beta,\text{curt}},\, D_{t,j,o}^{\beta,\text{bo}},\, \Delta S_{j,o}^{\alpha-},\, \Delta S_{j,o}^{\alpha+},\, \Delta S_{j,o}^{\beta-},\, \Delta S_{j,o}^{\beta+},\, G_{t,m,j,o}^{\alpha},\, G_{t,m,j,o}^{\beta},\, S_{t,j,o}^{\beta},\, S_{t,j,o}^{\alpha} >= 0 \tag{36d}$$

$$S_{(V+1),j,o}^{\beta} = s_{j,o}^{\text{init},\beta}, \quad S_{(V+1),j,o}^{\alpha} = s_{j,o}^{\text{init},\alpha} \tag{36e}$$

$$\text{for } \; t \in \{V+1,\ldots,V+24\},\, j,\, z \in J\text{: } P_{t,m,j,i}^{\text{r},-} = P_{t,m,z,i}^{\text{r},-},\, P_{t,m,j,i}^{\text{r},+} = P_{t,m,z,i}^{\text{r},+} \tag{36f}$$

$$\text{for } t \in \left\{V+1,\ \ldots,V+24\right\},\, j,\, z \in J \text{ with } \gamma_{t,j}^{\text{r},\pm} = \gamma_{t,z}^{\text{r},\pm}\text{: } \widetilde{P}_{t,m,j,k} = \widetilde{P}_{t,m,z,k} \tag{36g}$$

$$\text{for } \; t \in \{V+1,\ldots,V+24\},\, j,\, z \in J \;\; \text{with } \; \gamma_{t,j}^{\text{r},\pm} = \gamma_{t,z}^{\text{r},\pm},\, \psi_{t,j} = \psi_{t,z}\text{: } Y_{t,m,j,o} = Y_{t,m,z,o} \tag{36h}$$

$$\text{for } \; d \in \{2,\ldots,D\},\, t_1 \in \{V+24(d-1)+1,\ldots,V+24d\},\, j,\, z \in J,\, t_2 \in \{V+1,\ldots,V+24(d-1)\},$$
$$t_3 \in \{1,\ldots,24(d-1)\} \text{with } \psi_{t_2,j} = \psi_{t_2,z},\, \gamma_{t_2,j}^{\text{r},\pm} = \gamma_{t_2,z}^{\text{r},\pm},\, \epsilon_{t_3,j,l}^{\text{called}\pm} = \epsilon_{t_3,z,l}^{\text{called}\pm}\text{: } P_{t_1,m,j,i}^{\text{r},\pm} = P_{t_1,m,z,i}^{\text{r},\pm} \tag{36i}$$

$$\text{for } \; d \in \{2,\ldots,D\},\, t_1 \in \{V+24(d-1)+1,\ldots,V+24d\},\, j,\, z \in J,\, t_2 \in \{V+1,\ldots,V+24(d-1)\},$$
$$t_3 \in \{1,\ldots,24(d-1)\} \text{with } \psi_{t_2,j} = \psi_{t_2,z},\, \gamma_{t_1,j}^{\text{r},\pm} = \gamma_{t_1,z}^{\text{r},\pm},\, \epsilon_{t_3,j,l}^{\text{called}\pm} = \epsilon_{t_3,z,l}^{\text{called}\pm}\text{: } \widetilde{P}_{t_1,m,j,k} = \widetilde{P}_{t_1,m,z,k} \tag{36j}$$

$$\text{for } \; d \in \{2,\ldots,D\},\, t_1 \in \{V+24(d-1)+1,\ldots,V+24d\},\, j,\, z \in J,\, t_3 \in \{1,\ldots,24(d-1)\}$$
$$\text{with } \; \psi_{t_1,j} = \psi_{t_1,z},\, \gamma_{t_1,j}^{\text{r},\pm} = \gamma_{t_1,z}^{\text{r},-/2},\, \epsilon_{t_3,j,l}^{\text{called}\pm} = \epsilon_{t_3,z,l}^{\text{called}\pm}\text{: } Y_{t_1,m,j,o} = Y_{t_1,m,z,o} \tag{36k}$$

The objective function (36a) is to minimize the expected total cost, which is defined as the sum of the total cost of each scenario weighted by its probability of occurrence. The total costs consist of energy procurement cost $C_{t,j}$ minus revenues $R_{t,j}$ from the participation in the reserve market. Based on the fact that the auctions take place repeatedly and that the amount of money at stake is relatively small, we assume that the considered company is risk-neutral and consequently include no risk measures in the objective function. In Eq. (36b) we include all relevant constraints associated with the production process and the participation in the energy-only and reserve market. Binary and continuous variables are specified in Eqs. (36c) and (36d). Based on the stochastic process of activations $\left\{\epsilon_{t,j,l}^{\text{called}+/-},\, \zeta_{t,j,l}^{\text{called}+/-}\right\}_{t\in T_2}$, we can a priori determine the initial inventory conditions for time horizon T_1 using Eqs. (((1), (2), (6), (7), (24)–(26)) in combination with the machine schedule of the time horizon T_2. The initial inventory conditions are defined in Eq. (36e). Since the problem is a multi-stage decision problem, decisions on each stage have to be made before any information of further stages is revealed. Therefore, we introduce non-anticipativity constraints in Eqs. (36f), (36g), (36i), (36h), (36j) and (36k), which ensure that decisions on a specific stage are made without the consideration of future information. The information revelation and decision structure is depicted in Fig. 2. On the first stage, the decisions on bids in the reserve market for the first day are made. At this stage, no additional information is available and therefore the bids need to be equal for all scenarios, which is enforced by Eq. (36f). On the second stage, reserve market prices are revealed and based on this information bids for the participation in the spot market can be determined. However, if two paths in the scenario tree share the same history (have the same reserve market price realizations at the first day) they must share the same decision on spot market bids for the first day. This is stated in

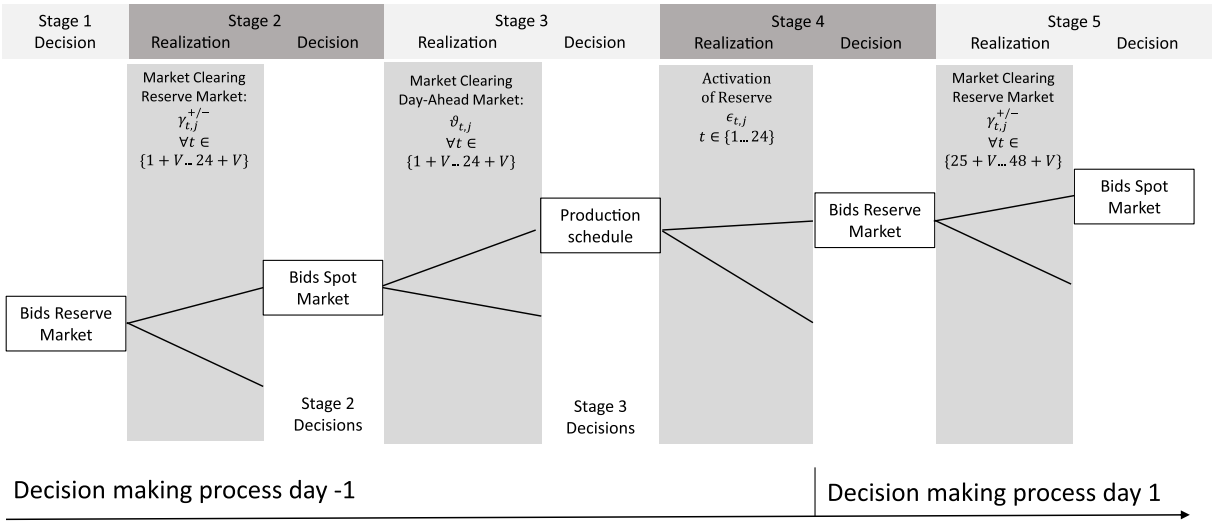


Fig. 2. Multi-stage decision structure and market participation.

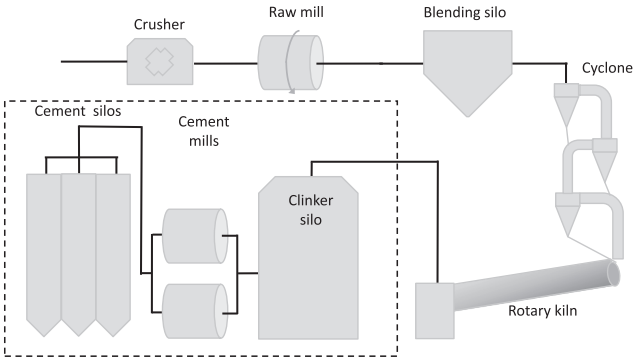


Fig. 3. Cement production process.

Eq. (36g). If the paths of two scenarios in the scenario tree share the same market prices in the first two stages, the same decision has to be made considering the assignment of products to machines for a specific time period. This is stated in Eq. (36h). The decisions on the subsequent days are constrained in a similar way. Eq. (36i) ensures that the same decision on reserve bid submissions is made for two scenario paths sharing the same history, including price realizations and activations of reserve of the preceding days. Eq. (36j ensures that the same decision on spot market bid submissions is made for two scenario paths sharing the same history, including price realizations of the spot market and activations of reserve of the preceding days as well as reserve market prices until the actual day. Eq. (36k) ensures that the same decision considering the assignment of products to machines is made for two scenario paths sharing the same history, including reserve and spot price realizations until the actual day as well as activations of reserve of the preceding days.

As stated in Section 2, the sequence of reserve and energy-only market can differ from country to country. Whenever the reserve market is conducted before the energy-only market, the non-anticipativity constraints need to be adapted to match the specific sequence of auctions. The adapted non-anticipativity constraints of the first day in this case read as

$$\text{for } t \in \{V+1,\dots,V+24\},\ j,z \in J:\quad \widetilde{P}_{t,m,j,k} = \widetilde{P}_{t,m,z,k} \tag{36f*}$$

$$\text{for } t \in \{V+1,\dots,V+24\},\ j,z \in J \text{ with } \psi_{t,j} = \psi_{t,z}:\ P^{\mathrm{r},-}_{t,m,j,i} = P^{\mathrm{r},-}_{t,m,z,i},\ P^{\mathrm{r},+}_{t,m,j,i} = P^{\mathrm{r},+}_{t,m,z,i} \tag{36g*}$$

$$\text{for } t \in \{V+1,\dots,V+24\},\ j,z \in J \text{ with } \gamma^{\mathrm{r},\pm}_{t,j} = \gamma^{\mathrm{r},\pm}_{t,z},\ \psi_{t,j} = \psi_{t,z}:\ Y_{t,m,j,o} = Y_{t,m,z,o} \tag{36h*}$$

The constraints of the subsequent days can be altered in a similar way.

4. Case study

4.1. Flexible operation of cement mills

As part of the project “Interflex4Climate” which was financed by the German Federal Ministry for Environment, Nature Conservation and Nuclear Safety the Karlsruhe University of Applied Sciences investigated a cement plant regarding its energy efficiency potentials and its capabilities to shift energy consumption. Within the project, a potential for demand response has been identified and a strategy for the production-inventory planning with participation in consecutive energy markets has been developed. The investigated company is located in North Rhine-Westphalia in one of the most important areas for limestone mining in Germany. The production process of cement is depicted in Fig. 3. In the first step, the raw materials, mainly lime marl, is quarried. The raw materials along with additives are grinded in the raw mill and stored in a blending silo. From here, the raw meal is fed to the multi-stage cyclone preheater. Subsequently, the hot meal is burned in a rotary kiln and cement clinker is produced as an intermediate product. The clinker is cooled down and stored in a clinker silo. In the final process step, the clinker is milled and three different main types of cement are produced. These differ mainly by granularity and grinding additives. The technical facilities of the company are controlled centrally and fully electronically and are therefore suitable for demand response. The investigation showed, that the production processes upstream to the kiln are mainly operated continuously. Flexible production units are the cement mills located between the clinker silo and cement silos. The plant is equipped with 2 mills with 2.5 MW installed electric power each. The approximate total annual energy consumption of the mills is 17.5 GWh/a. In Fig. 4 one of the mills is depicted. The production output of each mill is dependent on the type of cement. The inventory capacities of the cement silos are sufficient to store between 45 and 160 h of production output of the mills dependent on the given type of cement. Table 1 gives an overview on the key parameters of the production process.

4.2. Electricity market forecasting

Electricity price forecasting is a rapidly growing strand of literature in the field of energy forecasting with a high diversity of methods. Overviews and explanations of weaknesses and strengths of the forecasting tools can be found in the reviews of Weron [38] as well as Nowotarski and Weron [39]. Recent publications in the bidding literature focus on ARIMA and ARIMAX models to generate electricity price scenarios [22,28,20,23]. In the presence of high shares of renewable energies other methods that capture nonlinear effects of renewable production on electricity prices could be favorable. Since the implementation of a sophisticated forecasting model would exceed the scope of this work and spot price forecast are commercially available, we assume that we have a given spot price forecast and predict the forecasting error by a ARMA-model, similar as in [40,41]. The ARMA-series is defined as:

$$W^{\mathrm{err}}_t = a_1 W^{\mathrm{err}}_{t-1} + b_1 r_{t-1} + r_t \tag{38}$$

where W^{err}_t is the forecasting error of the spot market price in hour t and r_t are the innovations which are independent and identically distributed with $N(0, \sigma^2)$. This leads to forecasting error which increases with the forecasting time.

The investigation of the time series for German FRR indicates strong autocorrelation and significant correlations to the spot market price. Since the autocorrelation function (ACF) shows no sharp cut off we do not use moving average (MA) terms and represent the time series by



Fig. 4. Rotating cement mill.

Table 1
Process and cost parameter.

Parameter	Description	Value
b_1	energy intensity o = 1	21.925 t/MWh
b_2	energy intensity o = 2	25.25 t/MWh
b_3	energy intensity o = 3	13.24 t/MWh
$s_1^{\text{cap,max},\beta}$	inventory capacity o = 1	7,213 t
$s_2^{\text{cap,max},\beta}$	inventory capacity o = 2	5,376 t
$s_3^{\text{cap,max},\beta}$	inventory capacity o = 3	3,766 t
$c_o^{\beta,\text{curt}}$	demand curtailment cost	50 €/t
c^{su}	start-up cost	25 €
Δp_m^{ru}	ramp-rate of machine m	715 kW/min
$\sum_{t\in T_1} d_{t,1}^{\beta}$	Demand of product o = 1	3,936 t
$\sum_{t\in T_1} d_{t,2}^{\beta}$	Demand of product o = 2	3,589 t
$\sum_{t\in T_1} d_{t,3}^{\beta}$	Demand of product o = 3	1,269 t

ARX-models of the type:

$$\gamma_t^{r,+/-} = c\psi_t + \sum_{i=1}^{2} a_i \gamma_{t-i}^{r,+/-} L^6 + r_t^{+/-}. \tag{39}$$

Here, L is the back shift operator, c and a_i are the model parameters and $r_t^{+/-}$ are the residuals which are again assumed to be independent and identically distributed with $N(0, \sigma^2)$. These models capture correlations observed between spot and reserve prices.

To generate scenarios for the activation durations and start-ups in association with the participation in the reserve market we first model the reserve capacity demand within a specific time period and use historic merit order lists to calculate the market clearing price on a time grid with time intervals of 4 s. The autocorrelation function of the demand indicates significant autocorrelations for a large number of lags and the partial autocorrelation function cuts of after 2 lags. We, therefore, use AR(2)-models of the type

$$D_t^{r,+/-} = \sum_{i=1}^{2} a_i D_{t-i}^{r,+/-} + r_t^{+/-} t \in T_3 \tag{40}$$

to generate scenarios of the reserve capacity demand. The distribution of real FRR demand of one day and the distribution of the generated FRR demand are depicted in Fig. 6, additionally the left plot of Fig. 5 depicted one realization of the model. Even though we identified correlations between spot prices and supply curves, we did not predict reserve energy supply curves. Instead, we used historic supply curves to determine the market equilibrium for each time interval by the intersect of demand and supply. Whenever the simulated marginal market price exceeds a discrete price level l the corresponding bid is activated. Once a bid is activated/deactivated the set point is not modified for at least 10 min, this is in line with the prequalification requirements of the German TSOs [42]. For each hour of the optimization horizon and for each reserve demand scenario we calculate the expected activation durations and the number of activations of the discrete bidding price levels l. Fig. 5 illustrates the activation of a specific bid (bidding price = 90 Euro/MWh) during one hour.

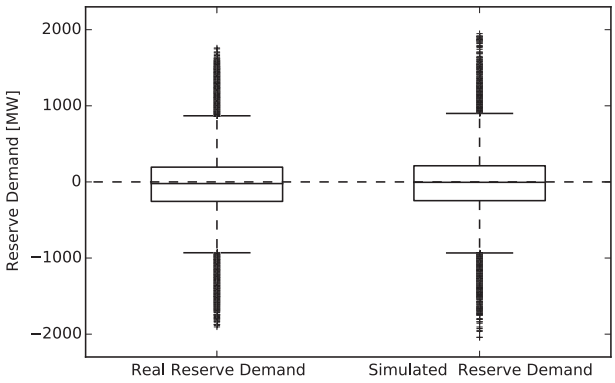


Fig. 6. Distribution of real and simulated reserve demand for one day.

4.3. Scenario generation and reduction

We represent the multivariate stochastic input process in the form of a scenario tree, which reflects the filtration of the process. Lately, the methods of scenario sampling and reduction have been applied in a stage-wise fashion [28,22]. These allow to preserve auto-correlations and cross-correlations of the time series and, furthermore reduces the memory usage as well as computational times of the scenario sampling and reduction procedure. For scenario sampling, we use the models described in Section 4.2, for scenario reduction we use the k-medoids clustering algorithm and as a distance measure the cumulative euclidean distance between price points of the scenario paths. We assume, that spot prices are independent of reserve prices and reserve activations and therefore first generate and reduce spot price scenario paths for each day conditional on the spot price scenario paths of the previous day. Subsequently, we sample conditional scenario paths of the reserve prices, activation durations, and the number of activations.

For the first day we sample $J_1^{\text{ini,spot}} = 100$ scenarios and reduce them to $J_1^{\text{red,spot}} = 6$ for each subsequent day we sample $J_2^{\text{ini,spot}} = 100$ scenarios for each scenario of the previous day and reduce them to $J_2^{\text{red,spot}} = 2$ spot price scenario paths. This results in a scenario tree with 48 scenarios which is depicted in Fig. 7, in which each line represents one spot price scenario. The market volume traded on the spot market is on average 20 times higher than the volume on the reserve market. Therefore, we assume that reserve prices are driven by expected spot

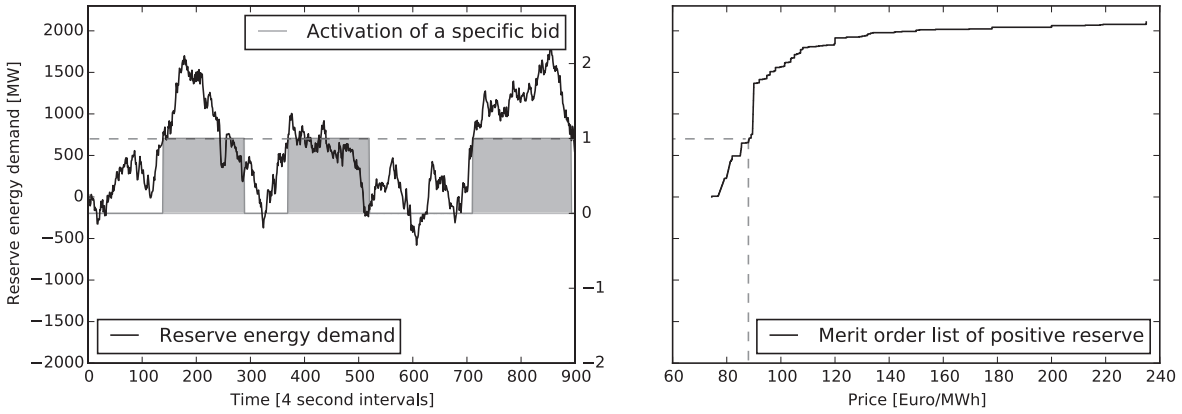


Fig. 5. Randomly generated reserve demand and resulting activation of specific bid according to the merit order (indicated by grey filling).

prices, even though reserve prices are revealed before spot prices. Given this information structure, we sample $J_d^{\text{sam,r}} = 6$ scenario paths of positive and negative reserve prices for each day and each spot price scenario path and reduce them to 1. Furthermore, we sample three reserve demand scenario paths for the first day ($J_1^{\text{ini,ra}} = 3$) and each spot price scenario path, for each subsequent day $d \in \{2, ..., |D|\}$ we sample $J_d^{\text{ini,ra}} = 1$ demand scenario paths for each scenario path of the preceding day $d - 1$. Given this procedure, we generate a scenario tree with 144 scenarios.

4.4. Further parameterization

We compare the energy procurement on the day-ahead and reserve market to a reference case, which is the energy procurement under a day/night energy tariff. To ensure comparability we use our day-ahead electricity price forecast to calculate the average electricity price during night and day hours and use these prices as night and day tariffs. This ensures that identified cost reductions are caused solely by load shifting and shaving and not by the avoidance of retailer margins. We use discrete bidding prices for spot and reserve market bidding. While we use static values for reserve energy bid price points $c_l^{\text{re},+/-}$, we calculate price points $c_{t,k}^{s}$ and $c_{t,i}^{\text{re},+/-}$ dynamically for each time point t, by the discretization of the distance between the price points of the two scenarios with the minimal and maximal value at time point t.

4.5. Investigated bidding strategies

We investigate four different energy procurement and capacity marketing strategies to illustrate the practicability of our approach. Under the first strategy, the production schedule is determined considering energy procurement and billing according to a day/night tariff. This represents the reference case and the current arrangement of energy procurement in the company. Under the second strategy, we determine market bids and the production schedule to minimize the expected energy procurement cost of the company when participating in the day-ahead market. This provides a solution of coordinated bidding in sequential energy-only markets, which has been widely investigated in the literature In the third strategy, we use our decision support tool to determine optimal bidding and production schedule for the participation in the day-ahead and reserve markets, anticipating uncertainties of market prices, contracted reserves and activation of reserve. This scenario enables us to, on the one hand, illustrate that our approach is capable to identify feasible solutions in a real-world setting and, on the other hand, provides insights on the potential of a coordinated bidding strategy in the German energy and reserve markets. Under the fourth strategy, we consider time-invariant bidding behavior at the reserve markets (i.e. firms submit identical bids at all reserve market stages) but optimally participate at the energy-only markets. In this strategy we determine the optimal time-invariant bidding behavior at the reserve markets allowing us to determine a lower bound on the cost decrease our newly proposed approach will provide, compared to a heuristic time-invariant bidding strategy on the reserve market in combination with sequential bidding in the energy-only markets.

4.6. Results

In the following, we present the results of the defined bidding strategies applied within a real world case study as an illustrative example of the proposed methodology. The top subplot of Fig. 8 depicts the committed capacities of the mills under a day/night tariff. It can be seen, that the production capacities are designed to meet the demand by primarily producing in base tariff hours. Since inventory deviation cost are evaluated by a cost factor, which is calculated by the product of anticipated future electricity price and the electricity intensity of each product, the machines are shut down whenever the electricity price exceeds the anticipated future electricity price. A detailed overview of start-up, energy procurement, and inventory deviation cost is given in Table 2. The middle subplot of Fig. 8 depicted the committed capacities with energy procurement on the day-ahead market for one selected scenario. The plot shows bidding prices (solid lines) as well as expected market prices (dotted line) of the selected scenario. Since the company is a buyer in the energy-only market, bids are accepted whenever the market price is below the bidding price. Production is shifted towards hours of low expected market prices. Since price fluctuations are exploited, the number of start-ups and the associated cost increase, while the energy procurement cost are significantly reduced. As illustrated in the cost overview in Table 2 the overall cost is reduced by approximately 13% while the consumed energy increases slightly and inventory deviations from aspired inventory levels decrease.

The bottom plot of Fig. 8 shows the energy consumption based on the energy procurement on the day-ahead market with participation in the reserve market for one selected scenario. It can be seen that the energy procurement on the spot market is significantly reduced and that the energy procurement on the spot market is shifted towards full 4-h time slices fulfilling requirements of the market participation in reserve markets. Activations of negative reserve can be observed relatively frequently in comparison to activations of positive reserve, this suggests that reserve energy bids of negative reserve are relatively low and reserve energy bids of positive reserve are relatively high. A detailed depiction of the sequential market participation in the day-ahead and reserve market is presented in Fig. 9. The top subplots show bidding price in the day-ahead spot market and resulting machine commitments. The second plot from the top indicates that, whenever a machine is planned to be committed on the spot market, the associated capacity is offered on the reserve market as well. The energy bids in the reserve market are chosen to be high to avoid frequent activations of reserve and associated cost, as indicated in the third subplot. Since activations in times of low day-ahead market prices yield a higher cost of activation, reserve energy bids are chosen to be higher during these time intervals. Bidding prices and associated committed capacities in the negative reserve market are depicted in the second plot from the bottom in Fig. 9. Energy bid prices are chosen to minimize the cost of activations while ensuring acceptance of the corresponding capacity bid. Since the activation of reserve leads to increased production output, the company is willing to pay for activations. This is reflected by negative energy reserve bid prices. The cost overview in Table 2 indicates that the total energy cost can be reduced by approximately 43% when participating both in day-ahead and reserve markets compared to the energy procurement under day/night tariff. While energy procurement cost on the day-ahead market increase slightly, the reserve markets offer significant potentials to gain additional revenues by the provision of reserve capacity and energy. These revenues outweigh the cost associated with the reservation and actual activation of capacity. Additionally, we investigated a heuristic participation in the reserve

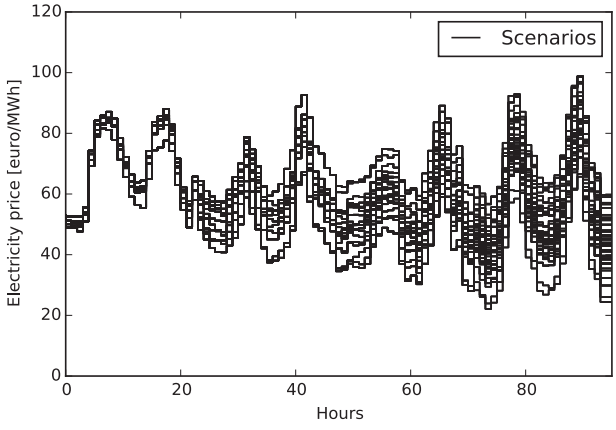


Fig. 7. Scenario tree containing 48 spot price scenario paths.

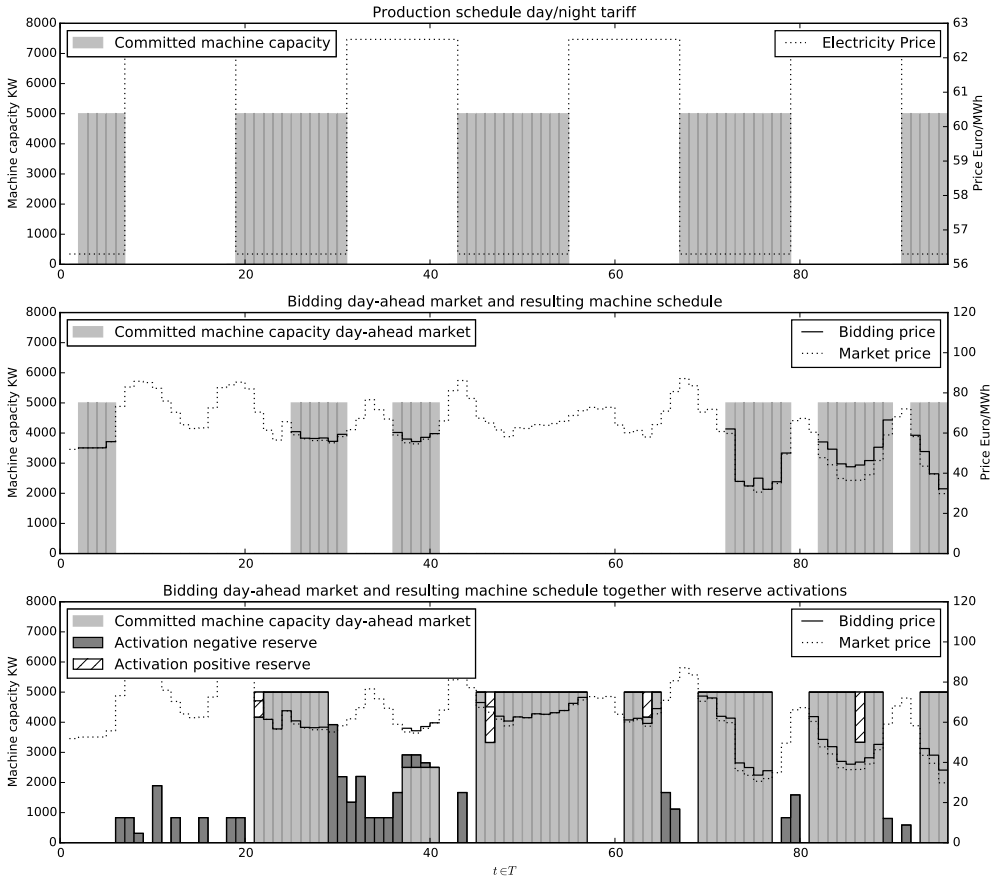


Fig. 8. Committed capacities with day/night tariff (top), with participation in the day-ahead market (middle) and with participation in the day-ahead market and reserve market (bottom).

Table 2
Average cost over all 144 scenarios for all procurement strategies.

		Day/Night Tariff	Spot Market	Spot and Reserve Market	Heuristic reserve market participation
Start up cost	[€]	150	201	150	129
Spot and Base/Peak procurement cost	[€]	13,230	12,644	13,520	12,386
Capacity renumeration revenues positive reserve	[€]	0	0	3,042	365
Capacity renumeration revenues negative reserve	[€]	0	0	2,335	937
Energy renumeration revenues positive reserve	[€]	0	0	894	49
Energy renumeration revenues negative reserve	[€]	0	0	−450	20
Start up cost reserve activation positive reserve	[€]	0	0	222	249
Start up cost reserve activation negative reserve	[€]	0	0	1,075	229
Inventory deviation cost	[€]	8,616	6,308	3,306	6,452
Total Cost	[€]	**21,997**	**19,154**	**12,453**	**18,094**
Cost Reduction compared to Day/Night Tariff	[%]	0%	12.93%	43.39%	17.74%

market in combination with a fully coordinated bidding in energy-only markets. Within this strategy, the company identifies cost-optimal time-invariant bids to participate in the reserve markets. These bids are identified by our approach by including additional constraints to enforce positive and negative capacity and energy bids in the reserve market to be equal in all scenarios and time periods. This allows us to determine the lower bound of the energy procurement cost that can be realized by a time-invariant separate bidding strategy. The results indicate that the optimal time-invariant bids consist of 2.5 MW in the positive and negative reserve market, respectively. The optimal time-invariant capacity prices (energy prices) are 73.78 Euro/MW (100 Euro/MWh) and 51.1 Euro/MW (12.5 Euro/MWh) for the positive and negative reserve market, respectively. The cost overview in Table 2 indicates that the total energy cost can be reduced by approximately 18% when participating both in day-ahead and reserve markets under a time-invariant strategy compared to the energy procurement under day/night tariff. That indicates that only 15% of the potential of the coordinated bidding strategy can be realized by the time-invariant bidding strategy, which could be a first heuristic approach to participate at both markets. The additional cost compared to the coordinated bidding strategy results from not anticipating the spot price trajectory when bidding in the reserve market. Resulting dispatch can either enforce energy procurement on the spot market at high price or impede energy procurement at low cost. Additionally, the activation of reserve as a result of the time-invariant bidding price can not be affected, resulting in higher storage deviation cost as illustrated in Table 2.

5. Conclusion

This work proposes a decision-support tool for the integrated

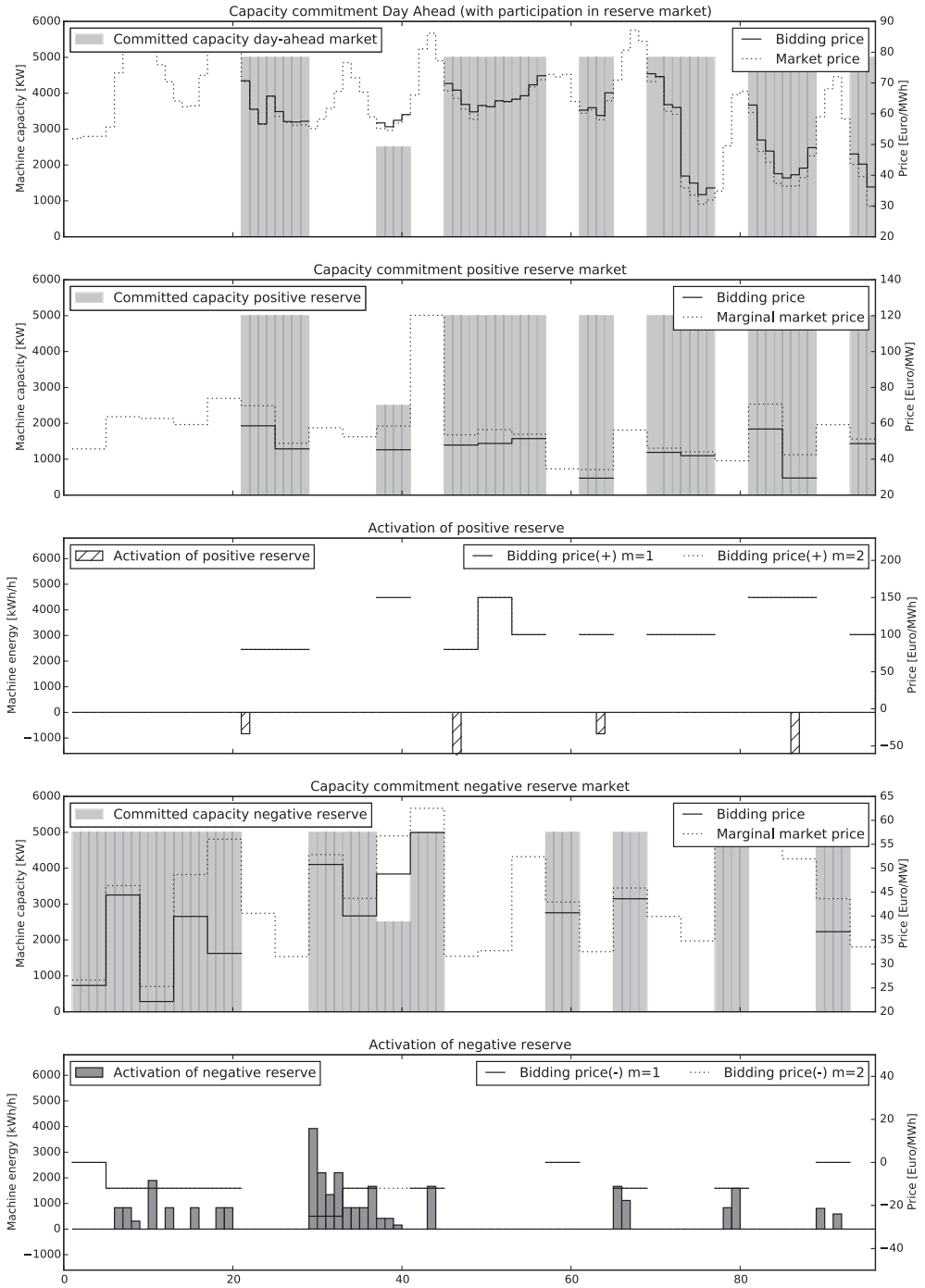


Fig. 9. Committed capacities day-ahead market (top), positive reserve (middle top), negative reserve (middle bottom) and expected activation of reserve (bottom).

production-inventory planning and electricity market participation of an energy-intense industrial plant. We consider a multi-product and multi-machine production scheduling problem and include decisions associated with the bid submission in sequential electricity markets. Information is revealed in a stage-wise fashion which results in a mathematical representation of the problem as multi-stage stochastic mixed-integer programming problem. We consider the energy-only market as well as reserve markets and the associated price uncertainties. We develop a new approach to account for the uncertain activations of reserved capacity, which allows us to transfer the continuous process of reserve activation into hourly stochastic processes. We further propose a flexible formulation of the reserve market model to ensure easy adaptability to different market characteristics such as scoring and pricing rules, activation times as well as auction frequencies and bid sizes. We conduct a real-world case study and apply our approach to the production planning process of cement plant participating in the German electricity markets. Reasonable computational times together with high economic potentials demonstrate the applicability of the presented approach. Within the investigated horizon, the German market offers significant activation incentives for flexible industrial loads to actively participate in the day-ahead as well as in the secondary reserve market. Compared to a time-invariant bidding strategy in the reserve market the cost can be reduced considerably by the proposed coordinated bidding methodology.

Declaration of Competing Interest

The authors declare no conflict of interest.

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